Lab # 01

Implementation of Relational Database Model using MS Access

Objective:

- Generate ERD to tables
- Generate tables in MS Access.

Software Tools:

- Microsoft Access

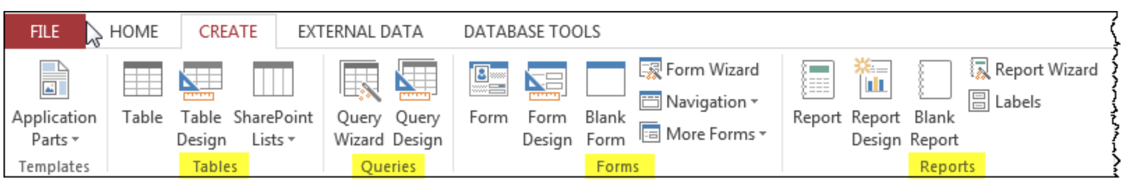
Theory:

Microsoft Access is a database management system (DBMS) from Microsoft that combines the relational Microsoft Jet Database Engine with a graphical user interface and software-development tools.

A Database Management System (DBMS) is a set of procedures and tools to store and retrieve information. The database itself is the stored information. The types of information stored in the database are defined by the corresponding data structures. The database structure overall consists of the tables, their constituent fields, and the relations between them. All information in the database is stored in these tables.

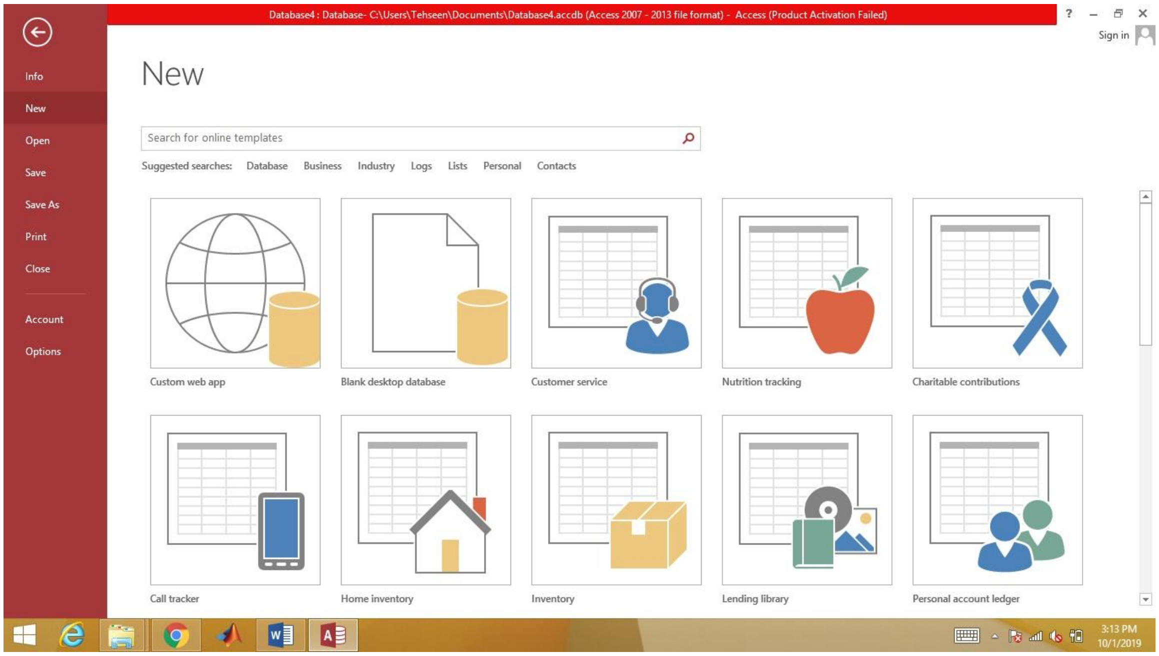
A DBMS consists of more than just the data. The DBMS also includes forms, queries and reports. The forms are the displays for screen and print that allow entering new information into the database tables and displaying the existing information. The queries are searches of the database that extract specified information. The reports are formatted displays of the extracted information, for screen and for print.

These four database constructs are available from the Create tab in Access.



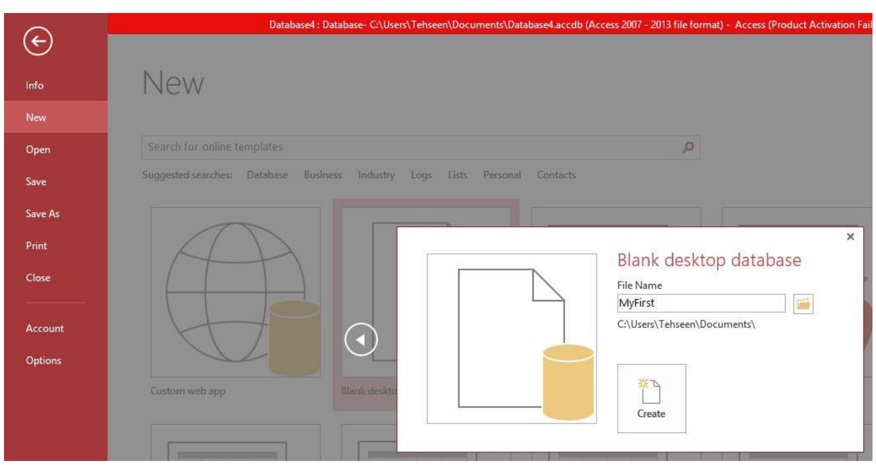
Following are the directions for creating a simple database to generate an invoice for selling products to customers.

1 Start Microsoft Access:



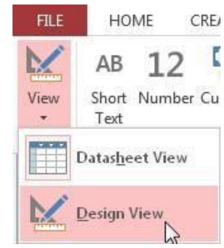
2 Create a New Database

There are two ways to interact with a database: design and use. The database user can be, and most often is, oblivious to the underlying design principles. However, someone has to design the database, to create the tables and their relations, to build the forms, and implement queries and reports. The designer interacts with the database in various design modes that the most users never experience. These interactions are typically iterative, as initial design are continually modified as a reflection of user experience. After starting the Access application to begin a new database click the blank desktop database icon, at the bottom right of the opening window, specify the name of this new, blank database here named **MyFirst.** Note where the database is stored, by default, inside the Documents folder. If for example, you are working on the public computer, you will wish to store the file on the system that you can access from anywhere such as on your drive H: drive. Click the **browse** icon to change the location. Click the **create** button to proceed.



3 Customer and Product Tables

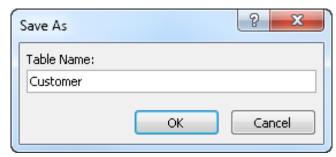
To construct an invoicing database, there needs to be customers who wish to purchase the products to sell. The first steps for the design of this database are the creation of the respective tables to hold the customer and product information.



Nothing can happen in a database without at least one table to store information. A new database opens with a table called Table 1, created and ready for additional modification.

A DBMS requires different views of the database from which to work. Each View accomplishes a different task, with its own functions, rules and menu system. The most fully featured editing of the database structure takes place in the Design View. To move to this View, click the View drop down menu at the top-left corner, select Design View.

2.1 Customer Table



2.1.1 Name the Customer Table

The initial database structure consists of only a blank table named Table 1, with no fields. When moving to the Design View a prompt appears to save the table, with an option to provide a new name. Call the first table in the Invoicing Database the Customer table.

2.1.2 Save the Changes

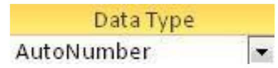
At any point, the changes made to a table or other database entities can be accomplished by rightclicking the corresponding tab. For the Customer table, right-click the tab named Customer, and select Save to save changes up to that point.

2.1.3 Primary Key

After naming the table, a window appears with the fields for the table listed by rows. The first field is the table's primary key field, which can be reset by clicking the Primary Key icon at the top left in the toolbar. The default Field Name of the primary key field is ID.

2.1.4 Data Type

Each field in a database stores information in a specific way, defined by its Data Type. Most data fields tend to be Short Text fields for generic text, but there are several other possibilities shown in the accompanying table. Change the data type by clicking the

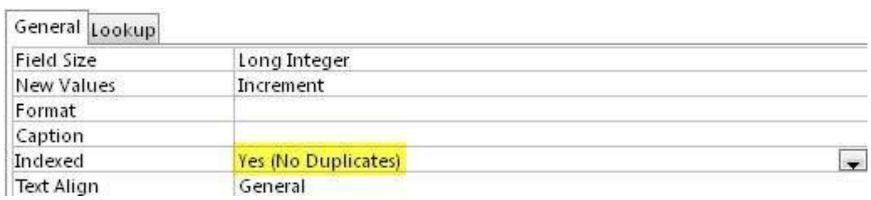


Data Type entry for a row and then select an option from the resulting drop down menu.

2.1 Customer Table

Data Type	Description
Text	Alphabetical/numerical data, up to a maximum of as 255 characters, such names, addresses, room numbers, zip codes, etc.
Memo	Up to 32,000 characters of text.
Number	Numbers for arithmetic calculations.
Date/Time	Dates and Times.
Currency	Dollars ($).
AutoNumber	An automatic counter that assigns a number each into a time data is entered new field.
Yes/No	Binary data, such as True/False or Yes/No.

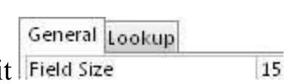
The primary key field has a default Data Type of AutoNumber. For the Customer table, the primary key field serves as the Customer ID. For a database to function successfully, each primary key field must possess specific properties, such as uniqueness. In terms of the Customer table, no two customers should have the same ID. This uniqueness property is assigned by default to primary key fields, as indicated by the No Duplicates term in the Indexed row of Field Properties.



2.1.6 Add Remaining Customer Fields

To add new fields to the table, just place the cursor in the first blank row in the first column, the Field Name column, and enter the name of the field. The Customer table should contain the following fields in addition to the Customer ID: FirstName, LastName, Addr, City, State, and Zip. To add the FirstName field, just enter in the name of the field and accept the default Data Type of Short Text.

2.1 Customer Table



The default Field Size of a Short Text field is 255 characters, a bit large for a first name. The standard length of a first name field is around 15 characters, so change accordingly. Now the database designer must make some decisions. Should a FirstName be required? If not, leave the required attribute set at the default No. Also, first name fields are generally not indexed because their values are generally not searched for and/or sorted.

Next enter the LastName field. Set the length of the Short Text field to approximately 30 characters. Because the operation of customer-based DBMS performs queries and sorting the last names, index this field, but do allow duplicates. Also, require this field to be present on any Customer record, with at least one character of content.

The remaining Customer table fields can now be entered, all as Short Text fields. No field need be longer than 30 characters, and the State field need only be 2 characters wide.

General | Lookup

Field Size	30
Format	
Input Mask	
Caption	
Default Value	
Validation Rule	
Validation Text	
Required	Yes
Allow Zero Length	No
Indexed	Yes (Duplicates OK)

Customer

Field Name	Data Type
ID	AutoNumber
FirstName	Short Text
LastName	Short Text
Addr	Short Text
City	Short Text
State	Short Text
Zip	Short Text

2.1.7 Enter Customer Data

The customer table has been created, so now customer data can be entered. In a production environment, data entry and modification are usually accomplished with forms, each form constructed to fulfill a specific purpose, such as entering Customer data. Access also provides a kind of "no frills", more direct means for entering data without constructing a form. To do this, move from the Design View to the Datasheet View, such as by clicking the View drop-down menu at the top-left under File, or by right-clicking the Customer tab and selecting the Data Sheet option. The Datasheet View lists the fields horizontally.

Customer

ID	FirstName	LastName	Addr	City	State	Zip
(New)						

Each row represents the data for a specific record. In this example, the ID field is set to the data type of AutoNumber, so no value is entered into that field. Begin by entering a customer's first name, and then continue for the rest of the fields for that customer. Data can be entered at any one time for as many customers as desired. Here data for two customers are entered.

Customer

ID	FirstName	LastName	Addr	City	State	Zip
1	Sam	Smith	44 NE 44th St	Portland	OR	97201
2	Sally	Surich	55 SW 55th St	Portland	OR	97231
(New)						

2.2 Product Table

2.1.8 Print Customer Data

Printing from Access is the same as from any application. Make sure the window is open and selected, with the data in view, here from the Datasheet view. Then, from the Office Button choose the Print option.

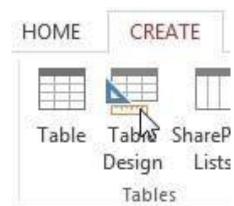
When finished entering customers, right-click the yellow Customer tab above the ID field and Save and then Close the window.

2.2 Product Table

Before creating an invoice, there has to be customers who wish to purchase the products the company has to sell. Next create the Product table and enter some products. Follow the same procedures outlined for the Customer table. First create the Product table, then enter the data in Datasheet View.

2.2.1 Create Table

To create a new table, go to the Create tab at the top-left of the Access window and click the Table Design button.



2.2.2 Define Fields and Primary Key

A Product table includes a Product ID that uniquely identifies each product. Within the database, the Product ID serves as the Product table's primary key. In the first row of the table as viewed in Design View, enter ProductID as the field name and select AutoNumber as the Data Type. To set as the primary key, rightclick in the cell for the Field Name and select the Primary Key option.

2.2.3 Add Description and Price Fields

To complete this minimal product table, also include a Description field, which contains a word or brief phrase that describes the product, as well as the selling Price to appear on the invoice. Keep the default Data Type of Short Text for the Description field, but limit to about 30 characters. Set the Data Type for the Price field as Currency.

2.2.4 Add Products

Again, switch to Datasheet View as was done with the Customer table on page 4. Save the table with the name of Product. Then add some products to the company store.

ProductID	Description	Price	Click to Add
1	Jacket Liner	$199.00	
2	Pant Liner	$199.00	
3	Gloves	$139.00	

2.2.5 Print Product Data

Printing from Access is the same as from any application. Make sure the window is open and selected, with the data in view, here from the Datasheet view. Then, from the Office Button choose the Print option.

3 Database Structure

3.1 Order and OrderLine Tables

The minimal structure of an invoicing database contains four tables, two more in addition to the already created Customer and Product tables. One of these tables is the Order table, which stores the individuals orders. Another table is the OrderLine table, which contains individual products ordered as well as the corresponding quantity for each order. However, unlike the previous Customer and Product tables, create the tables but do *not* enter data directly into the tables. Instead, an Invoice form will be created, and only through this form will data be entered into the Order and OrderLine tables.

3.1.1 Create the Order table

Create the Order table and add the respective fields to each table. The Order table contains the primary key OrderID as well as the OrderDate. Set the primary key for the table. The data type for OrderDate is Date. Right-click the table tab above the Field Name field and Save the table. When prompted, name the table Order.

3.1.2 Add the Order Table Foreign key

The Order table also contains a foreign key that links a selected order with the customer that placed that order. Name the foreign key fkCustID and assign the same attributes as the corresponding primary key CustID in the Customer table except that a foreign key is not unique. In this case, one customer will hopefully have many invoices. A useful convention places the primary key in the first field and any foreign keys in the immediately adjoining fields.

> **Caution** → The attributes of a foreign key match the attributes of the corresponding primary key except that the content of a foreign key is *not* Unique.

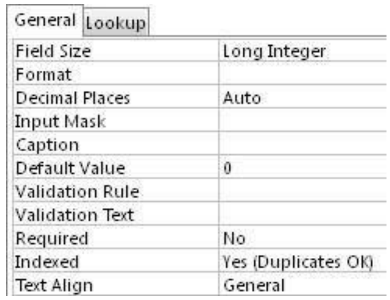
Since the corresponding primary key has a Data Type of AutoNumber, set the Data Type of the foreign key to Number.

Order

Field Name	Data Type
OrderID	AutoNumber
fkCustID	Number
OrderDate	Date/Time

3.2 Table Relations

Also, make it required and indexed, though with duplicates allowed.



3.1.3 Save and Close the Order Table

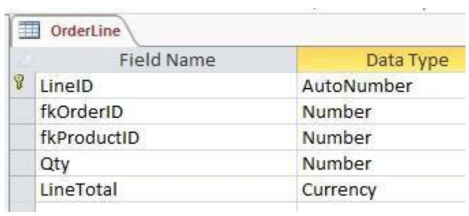
Right-click the tab now named Order and once again Save the data. Only this save is necessary, however, as with any computer application, it is advisable to save often, not just when the entered information is complete. After saving, Close the window. Generally, when the immediate task for a window is completed, the window should be closed to avoid cluttering the interface and to make sure everything is saved and out of the way before continuing with the database operation as a user.

3.1.4 Create the OrderLine Table and Fields

Follow the above procedure for the OrderLine table. The OrderLine table is a many-table to both the Order table and the Product table, which requires two foreign key fields. These foreign key fields respectively link each OrderLine record to a specific Order as well as to a specific Product. Again, the foreign key fields are required and indexed, but with duplicates.

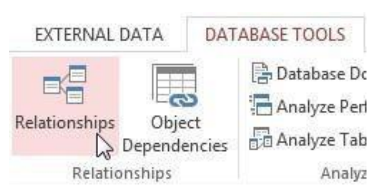
The Data Type of the LineTotal field is Currency.

There should also be a primary key for this table, even though it is not used directly in the database, and is not shown in the following figure. Name the primary key field something such as LineID, and assign a DateType of AutoNumber.



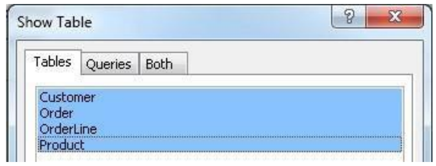
When finished, Close the window that contains the OrderLine table. Again, do not enter any data, a task accomplished with the Invoice form to be constructed later.

3.2 Table Relations

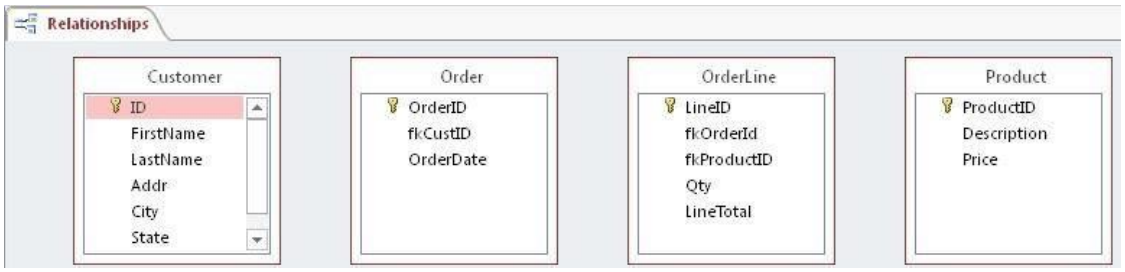


A crucial element of database structure is to relate the tables with the appropriate one-to-many relationships. To accomplish this, click the Database Tools tab at the top of the screen and then click the Relationships button.

Then select all four tables in the resulting Show Table window. To do this, click on the first table, Customer, then hold the Shift key down and click on the last table in the list, Product. The click the Add button.

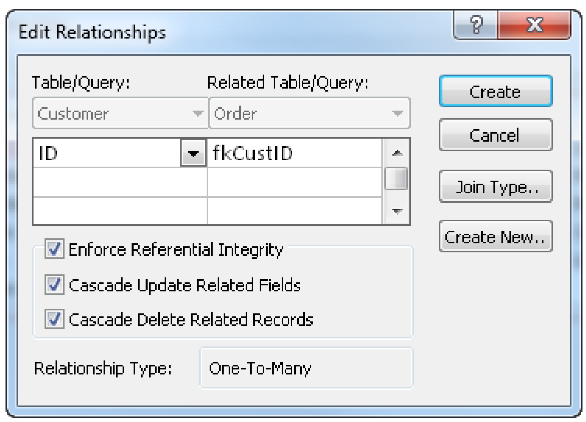


The result is all four tables and their fields displayed in the Relationships window.



Now relate the tables with their corresponding one-to-many relationships. Click the Customer ID field, hold the mouse down and drag to the foreign key for Customer ID in the Order Table. Release the mouse and the Edit Relationships table appears. Check the

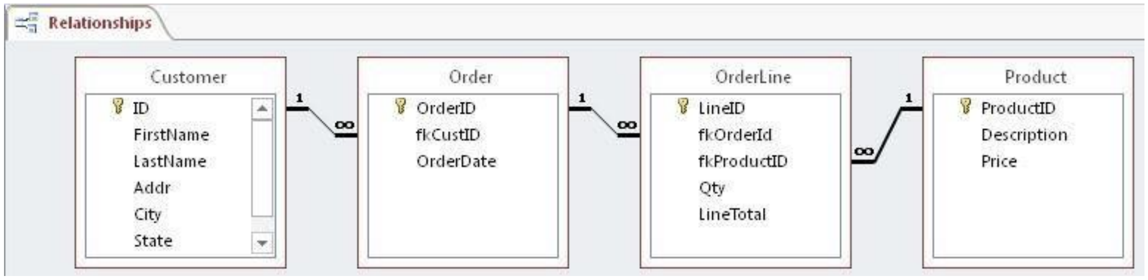
Enforce Referential Integrity as well as the Cascade Update and Cascade Delete checkboxes. Then click the Create button.



Similarly, define the one-to-many relationships that related the Order table to the OrderLine table and the Product table to the OrderLine table. Relating all four tables results in the following database structure, as shown in the Relationships window. This figure illustrates the Access representation of the general invoicing database structure.

Caution → Check that your database structure *exactly* matches this structure in the accompanying figure before continuing.

These four tables and the corresponding three one-to-many relationships define the minimal structure required to produce a working invoicing database. Each of the four tables



should contain the fields specified in the preceding figure. When complete, Close the Relationships window.

3.2.1 Print the Database Structure

To print the database structure diagram, under the Database Tools ribbon, make sure the Relationships button is pushed to display the data base structure with the relationships and tables displayed. Then go to Design ribbon which appears when the relationships are displayed and select the Relationships Report button. Click the button, which causes the database structure diagram to appear in a window ready to print. Then print, such as to a pdf, as usual. To print a pdf, choose the PDF or XPS button toward the top-right of the File ribbon.

Lab Task:

A. Design a database system for a library inventory system. The library recently received a large collection of books and would like to keep better track of the titles. The library is still using an antiquated method for the cataloging most its books. Because of your computer skills, you have been asked to create a database containing the catalog number, title, author and copyright date of each of the new book. After the database is created, the database will make search, tracking and inventory functions much easier.

Catalog #	Title	Description	Acquired Date	Location	ISBN	Condition
1	Speak	Paperback	8/12/2006	Building	340817623	Binding loose
3	Hoot	Paperback	8/17/2006	Building	375821813	
4	Artemis Fowl	Hardcover	8/17/2006	Bookmobile	141312122	Missing dust jacket
5	Wolfie	Hardcover	9/29/2008	Bookmobile	299389231	
6	Student Name	Paperback	9/29/2008	Bookmobile	555555555	

B. Create the table from this ERD

Vehicle	
Vehicle Type	Short text
Reg_No	AutoNumber
Model	Short text
Dealer	
CNIC	AutoNumber
Name	Short text

Phone	Number
Picture	Attachment
Mechanic	
Name	Short text
CNIC	AutoNumber
Address	Memo
Name	Short text
Charges	Currency